Six-Sided Dice Layer Cake

A memory-first machine for language model inference — concept brief

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In one paragraph

Running a language model is a memory problem wearing a compute problem's clothes. A modern accelerator has enormous arithmetic and spends most of it waiting for weights to arrive. The industry's answer has been to stack memory closer to the processor. This design inverts the arrangement instead: a sixty-millimetre cube with six processors on its six faces, all looking inward, and at the geometric centre a solid block of static memory holding every weight of the model. The weights never move. There is no DRAM, no memory bus, and no off-chip traffic for weights at all.

The trade, stated plainly

Against a current accelerator with stacked memory, this design's own model puts it at **11.4 times the memory bandwidth at 0.51 times the capacity.**

That is the whole proposition. Half the memory, eleven times the speed of reading it. It is a machine for a model that fits, run very fast — not a machine for the largest model that exists.

It is also explicit about its edge. Past its capacity the machine refuses rather than degrading: it offers a shorter context instead of quietly paging and getting slow. A hard ceiling is a narrower market and an honest one.

What it is, physically

A cube sixty millimetres on a side, weighing a little over a kilogram wet.

Six compute faces, one per side, each owning a consecutive run of the model's layers. A token falls through the six of them in sequence, the way grain falls through a stack of sieves. There are no face-to-face wires; everything passes through the centre.

The centre is forty millimetres of solid state: twenty-four tiers of static memory, laminated with twenty-four copper-molybdenum plates that carry the heat out sideways. Coolant enters at four corners of the cube and leaves at the other four, which is not a convenience — it is the only arrangement in which no supply channel joins two feed points, and every one of the twelve edge channels carries flow toward a load.

One of the six faces is not a processor. It is spent entirely on wire: sixteen million conductors, so that whatever the centre is holding can be somewhere else in thirty-four microseconds.

The name is the shape rather than a joke. A layer cake is the core — twenty-four tiers laminated with twenty-four plates — and a six-sided die is the six faces. A cube has three pairs of opposite faces, and a token visits all six in order: three of its five steps land on the face opposite the one just used, which is the most a cube permits, since an ordering has to cross between opposite-pairs twice and a crossing is always to a neighbour.

Specification

Every figure below is derived from the design's own geometry and physics rather than measured, and is regenerated from the blueprint set rather than typed.

Physical

edge	60 mm
volume	216 cm^3
mass, wet	1.13 kg

Power and cooling

input	48 V at 39.4 A
dissipation, design point	1891 W

dissipation, idle with a model resident	248 W
coolant	water, 3.52 L/min at 0.46 bar
inlet temperature	298 K
junction, hottest point, worst-served face	319 K
margin to the silicon's limit	59 K

Memory and compute

usable capacity	71.9 GB
largest model resident	36.4 GB of weights
aggregate read bandwidth	38.4 TB/s
arithmetic, eight-bit	4404 Tflop/s
model load, from storage	34.1 ms

Performance

tokens a second, one sequence	1021
tokens a second, aggregate at the design batch	19490
prompt tokens a second	25595
whole core out through the wire face	34.3 us

Note the ratio between the two token rates. Single-sequence and aggregate are about twenty apart, and that governs how this machine should be used. Below a batch of roughly twenty it is latency-bound and the arithmetic idles; above it, the machine is saturated and every additional sequence is nearly free. It is a serving machine, not a workstation.

The hard part, and why it is solved

Nineteen hundred watts inside two hundred and sixteen cubic centimetres is the problem that decides whether any of this is real.

It is removed by microchannel fields etched into six silicon cold plates, one per face, fed and drained through the eight corner blocks. The cold plates are silicon rather than copper so that they expand at the same rate as the dies bonded to them; the choice buys exactly zero expansion mismatch at that interface.

Two results are worth an investor's attention because they are counter-intuitive and they de-risk the design.

The cooling could also be made to slosh rather than circulate. As designed, coolant enters four corners and leaves the other four in a continuous loop, and every face sees fluid at the same inlet temperature. An alternative drives the column back and forth with a pump at each end: while one end is outside losing heat to a radiator the other is inside picking it up, and then the stroke reverses. It needs one channel per rail rather than two, which is the largest single reduction available to the cube's edge — and every dimension in this design hangs off that edge. What it costs is that the junction temperature stops being level and becomes a ripple at the stroke frequency. Which of the two wins turns on the ripple amplitude against the fifty-nine kelvin of margin, and behind that on the fatigue cost of cycling every bond in the machine at the stroke rate for ten years. It is recorded as an open question rather than settled.

Halving the coolant flow does not halve the cooling. In laminar flow the convection coefficient does not depend on velocity at all — only the coolant's own temperature rise doubles, and that rise is a fifth of the chain. A pump at half speed costs about four kelvin of junction temperature out of fifty-nine of margin. Partial pump failure is therefore graceful rather than catastrophic, and the redundancy design is built on that.

The coolant divides exactly evenly between the six faces. Not nearly — exactly. Of the sixty-four legal ways to connect twelve coolant rails to six faces, sixteen have a threefold rotational symmetry about a body diagonal that makes all six faces equivalent, and the flow through them cannot differ. The other forty-eight starve one face by five or six per cent. That was found by solving the fifty-branch hydraulic network for every legal arrangement, and the thermal budget is nonetheless built on the *worst* legal arrangement, so the design does not depend on the plumbing being assembled to the drawing rather than merely to the rules.

Where it wins and where it loses

Against one accelerator with stacked memory. Blank cells are figures this project does not hold and will not invent.

	This design	Comparator	
memory bandwidth	38.4 TB/s	3.4 TB/s	11.4 times
memory capacity	71.9 GB	141 GB	0.51 times
tokens/s, one sequence	1,021	—	latency
tokens/s per watt	10.3	—	density
tokens/s per litre	90,231	—	density
model load, from storage	34 ms	seconds	win
runs an arbitrary program	no	yes	**lose**
degrades past capacity	refuses	pages	differs
cooling	liquid, required	air or liquid	**lose**
software ecosystem	none	mature	**lose**

Four of the ten rows go against this design and one of the four — the software ecosystem — is the one that most often decides hardware outcomes.

The three unshared cases. What this does and a general accelerator cannot: serve a model that fits, at very low latency, in a fraction of the volume and power, because the weights are already where the arithmetic is. What a general accelerator does and this cannot: run anything else at all — training, a larger model, a different framework, a workload nobody anticipated. And where they are simply different: at the capacity edge, where one pages and slows and the other declines.

The latency advantage is structural rather than tuned. A conventional part fetches weights from stacked memory every token; this one does not fetch them.

How long the advantage lasts

The comparator improves. Assuming its memory bandwidth grows at a compound annual rate, the bandwidth lead closes after:

assumed annual improvement	years to parity
20 per cent	13.3
30 per cent	9.3
40 per cent	7.2

These are not forecasts and this project does not own them. The growth rates are assumptions declared in the design so that a claim about the future is a number a reader can change rather than a sentence somebody wrote. Substitute your own rate and the answer recomputes; the arithmetic is one line.

What the design asserts is the shape rather than the value: even against a comparator improving faster than stacked memory has sustained, the bandwidth lead outlives any plausible time to build one. **The capacity deficit does not close from this side at all** — the comparator is ahead there now and improving too, and that is the permanent half of the trade.

What it costs to run

The figures a buyer optimises are not the ones a user notices. Dividing the design's own numbers by each other:

tokens per second, per watt	10.3
tokens per second, per litre	90,231
tokens per second, per kilogram	17,303
gain from serving many sequences at once	19 times

The per-litre figure is the one the geometry earns. A conventional part spends most of its volume getting memory near the processor; this spends its volume on being memory near the processor.

The last row is the honest caveat. **Serving a single user, this machine wastes most of what it is.** It wants load, which makes it infrastructure rather than a desktop product, and narrows who it is for.

If the arrangement is better, why does it not exist?

This is the sharpest question available and it has three answers, none of which is comfortable.

Static memory costs far more per bit than DRAM. The trade only pays if you want bandwidth more than capacity. That is a bet about what inference workloads look like — the central bet in this design, and not a fact it rests on. If capacity turns out to matter more than bandwidth, this is the wrong machine and no amount of engineering rescues it.

Nineteen hundred watts in two hundred and sixteen cubic centimetres cannot be air-cooled. It needs liquid driven through the structure, so cooling is not a component bolted on afterwards — it is the frame, and it constrains every other decision in the design.

Nobody has a line for this. Six faces bonded around a sealed liquid loop, with a nine-step bonding sequence that has to run in descending temperature order, is a packaging problem rather than a silicon one. Finding out what such a line would cost is a large part of what the next step buys.

Those are the barriers and they are the only defensibility identified so far. **No patent position is claimed.** Difficulty is a weaker moat than a patent and a real one, and it is worth knowing which of the two is on offer.

The design as an asset

This is the part that is unusual and is worth understanding before valuing anything else.

The design is not a document describing a machine. It is written in a notation a program reads. Every dimension is either a number a person chose — there are eleven of those — or an expression over numbers that were. Alongside them sit **573 engineering requirements**, each carrying the author's own sentence saying why it must hold, and a checker evaluates every one of them in about a second.

All 573 currently hold. None of the design's own goals is unresolved: the notation has a category for a number the design wants and cannot yet produce, and the checker refuses to call the set finished while one exists.

For diligence, this means an engineer does not have to take the specification on faith. They can change an assumption and watch what breaks.

What that has already caught

Every one of these was found by the checker rather than by review, and each would have been an expensive discovery later:

- a corner manifold block that could not physically contain the chambers drawn inside it
- a die power grid four times too thin, found by the voltage droop it implied
- a bonding step hotter than the bond it was standing on, which would have reflowed the joint underneath it during assembly

- a memory stack that had to fall from thirty-two tiers to twenty-four once somebody derived the bitcell density instead of choosing a round number
- twenty-seven hand-written unit conversions inside derivations, three of which had produced visibly wrong answers and one of which was silent because two errors cancelled
- a cooling plate material chosen specifically for its thermal conductivity, in a design that had never once read that number
- a table of the six faces claiming every step of the pipeline landed on the opposite face, which is not a thing a cube permits — found only when somebody drew it, because an ordering is a list and the notation holds numbers

What is not known

Stated directly, because it is the part that decides what the next spend is.

Nothing has been fabricated. Not a die, not a cold plate, not a coupling. Every figure in this brief is derived from geometry and physics. That is a stronger position than a slide deck and a weaker one than a prototype.

There is no price. The bill of materials gives cost as ratios — silicon is 77 per cent of it, and memory tiers are two thirds of the silicon — and deliberately refuses a currency figure. With no volume, no supplier and no year, a number would be fiction.

Yield is the exposed flank. A memory tier is large enough that only about sixty-four per cent of them come off a wafer good. The design's own note is that making the tiers smaller would save more than any assembly improvement, and that nobody has yet asked whether a tier has to be one piece.

There is no software. A general accelerator arrives with fifteen years of tooling. This design assumes two pieces that do not exist: something to pack a trained model into the core's format, and a reference implementation to check the first generated token against, bit for bit. It is a machine that runs one shape of workload, so it needs far less tooling than a general part — but far less is not none, and nobody has written either piece. **This is the risk that most often kills hardware companies, and it is not an engineering problem.**

Two decisions remain open. Whether the coolant is water or a dielectric: water cools about ten times better, and a leak with water across a hundred and sixty-six joints is a dead cube rather than a mess. The substitution costs about eleven kelvin of the fifty-nine available, so the design survives either — which makes it a reliability judgement for whoever owns the failure consequences rather than a thermal one. And whether the model it is sized around is the right anchor, since that model sets the core, which sets the cavity, which sets the cube. A model half the size would let the whole machine shrink, and nobody has run that backwards.

Seventeen further questions are carried openly, each naming the part of the design that owns it and what it would change.

What it is ready for

An engineering review, not a fabrication run.

The most valuable next expenditure is somebody with process and packaging experience reading the fabrication and assembly sections and saying which of the 566 requirements they do not believe. The design is built to make that a cheap conversation: every requirement names its own reason, and changing an assumption and re-running takes a second.

All figures generated from the blueprint set. Regenerate with `./run-checks` *for the requirement status and* `luajit src/097-spec-report.lua` *for the specification. Nothing in this brief is entered by hand except its prose.*